

Question:

A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

- A. Autoimmune hemolytic anemia
- B. Microangiopathic hemolytic anemia
- C. Paroxysmal nocturnal hemoglobinuria (PNH)
- D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

<tool_call> reasoning process here

The patient is a 35-year-old African American man with a recent history of anti-malaria prophylaxis prior to travel to Nigeria. He presents with fever, abdominal pain, severe weakness, jaundice, generalized pallor, and laboratory findings of bite cells and Heinz bodies on peripheral smear.

Key points :

- Bite cells and Heinz bodies are classic findings in hemolytic anemia due to oxidative stress.
- Normal G6PD levels are important : G6PD deficiency typically presents with low G6PD activity, but here it is stated that G6PD levels are normal, which makes G6PD deficiency less likely.
- The patient has a recent exposure to anti-malaria drugs (such as primaquine), which are known triggers of hemolytic anemia in individuals with G6PD deficiency. However, the G6PD levels are normal, suggesting that the hemolysis is not due to G6PD deficiency per se.
- Bite cells and Heinz bodies are indicative of intravascular hemolysis due to oxidative damage, which is commonly seen in drug-induced hemolytic anemia, especially in the setting of malaria prophylaxis.
- The presence of these findings, combined with the recent exposure to a drug known to cause hemolysis in susceptible individuals, suggests a drug-induced hemolytic anemia. However, among the listed options, the only one that fits the clinical picture of oxidative hemolysis with bite cells and Heinz bodies is G6PD deficiency, but the G6PD level is normal.

Wait : This is a critical point. The question says "normal glucose-6-phosphate dehydrogenase (G6PD) levels." This is a key clue. If the levels are normal, then the patient does not have G6PD deficiency. Therefore, D is incorrect.

Now consider the other options :

- Autoimmune hemolytic anemia (A): Can present with jaundice and hemolysis, but typically shows spherocytes on peripheral smear, not bite cells or Heinz bodies. Also, no mention of autoimmune markers or recent exposure.
- Microangiopathic hemolytic anemia (B): Can have schistocytes and hemolysis, but bite cells and Heinz bodies are not typical. Also, no evidence of thrombotic microangiopathy (e.g., renal involvement, thrombocytopenia).
- Paroxysmal nocturnal hemoglobinuria (C): Presents with hemolytic anemia, thrombocytopenia, and often a history of hemolytic episodes. It can cause Heinz bodies and is associated with hemolysis, but it is not typically associated with bite cells or drug exposure. Also, it is more common in younger patients and has a different pattern.

Now, the key is : the patient has bite cells and Heinz bodies, these are classic signs of oxidative hemolysis, which is typically due to G6PD deficiency. But the G6PD levels are normal. So why is this happening?

Wait, a common mistake : "normal glucose-6-phosphate dehydrogenase (G6PD) levels" this is ambiguous. In clinical practice, "levels" may refer to the enzyme activity, not the gene. But G6PD deficiency is a functional deficiency. If the enzyme activity is normal, then G6PD deficiency is ruled out.

However, the question says "normal glucose-6-phosphate dehydrogenase (G6PD) levels" if this means normal enzyme activity, then G6PD deficiency is not present.

But the presence of bite cells and Heinz bodies is classic for oxidative hemolysis, which is the hallmark of G6PD deficiency, especially in the setting of drug exposure.

Is there a possibility that "normal levels" refers to a laboratory test that measures the enzyme, and the patient is truly normal?

Yes. So if G6